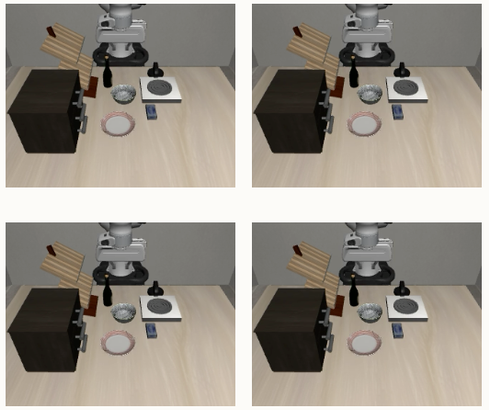
Robot demonstrations



“Put the bowl on the plate”

RGB + robot state + actions

Synchronized episode

Episode preparation & annotation

Standardization



Embodiment & coordinate mapping

Base frame **Gripper** **Joint order**

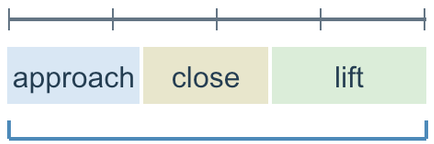


Pose units | Open / close | Joint mapping

Atomic Segmentation



start t contact finish c



One pick primitive

Completion event defines c

Clip to Typed Call

1 Infer operation from state changes

approach + close + lifted object = pick

2 Bind references and contract

scene: bowl arm: right result: held

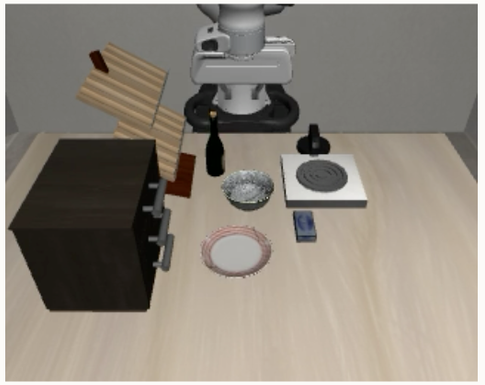
3 Serialize validated string slots

```
pick(object="bowl",
 target="above_table",
 arm="right", completion="held")
```

Ambiguous evidence goes to review

Endpoint geometry + event keyframes replace the function body

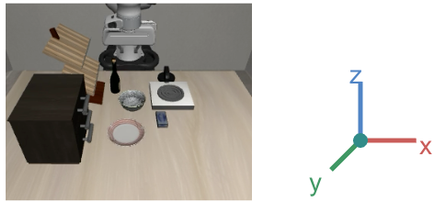
Current observation



t = start of this training window

Split episodes before sampling

Terminal Pose



End state: object held

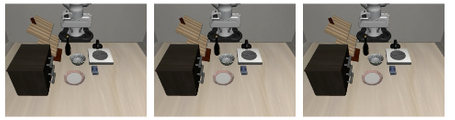
Query: (p^f, R^f)

Where the primitive finishes

One pose per arm

Event Keyframes

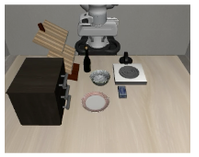
pick: pregrasp / close / lift



pregrasp closed lifted

move: endpoint

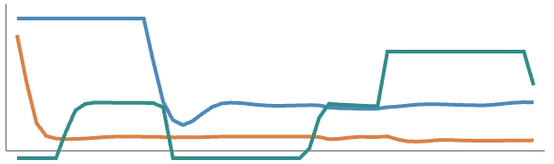
Different atoms, different events



arrived

Variable event count

Action Alignment



Actions realize event progression



valid actions mask boundary

$\{o_t, s_t, P^{txt}, X^f, l^f, A_t\}$

Event roles, timestamps and validity masks